

Applied Thermodynamics (ME2002)

Date: November, 2025
Course Instructor(s)

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Sessional-2 Exam

Total Time: 1 Hours
Total Marks: 40
Total Questions: 02

Semester: F-2025
Campus: Lahore
Deptt: Electrical
Engineering

Student Name

Roll No

Section

Student Signature

Vetted by

Vetter Signature

Instruction/Notes:

- Submit the question paper along with your answer sheet.
- Use valid assumptions if needed, don't ask any questions.
- Your answers should be correct up to three decimal places with appropriate units.
- Values read from tables should have the same number of significant figures as mentioned in the tables.
- You are allowed to bring two A4 sized cheat sheets, as well as thermodynamic tables

CLO #2: *Analyze energy of a system using energy equation*

Q1:

[marks 20]

Analyze energy of the following system. A steam turbine receives steam from two boilers. One flow (designated as A) is 5 kg/s at 3 MPa, 700°C, 20m/s and the other flow (designated as B) is 15 kg/s at 800 kPa, 500°C, 10m/s. The exit state is 10 kPa, with a quality of 96%. Find the total power out of the adiabatic turbine. What shall be percent change in output power if flow A is also supplying 15 Kg/s, keeping everything else constant?

CLO #3: *Apply Second Law of Thermodynamics to quantify the performance of a control mass and control volume thermodynamic system.*

Q2:

[marks =20]

Apply Second Law of Thermodynamics to quantify the performance of the following system. In a Carnot engine with ammonia as the working fluid, the high temperature is 60°C and as Q_H is received, the ammonia changes its quality from 10% to 90%. The heat rejection temperature is -20°C. Draw the T-s diagram, find the cycle thermal efficiency, the heat added per kilogram, specific work, the entropy and quality at the end of the heat rejection process. What should be mass flow rate to get an output power of 10 KW.